

Who Can Be Scored? Cohort Availability Limits Population-Level Capture in Prenatal Risk Triage

Anonymous Author(s)

Abstract

Clinical risk models are often evaluated only among individuals for whom a prediction can be generated, obscuring events outside the scoreable cohort. Using U.S. CDC Natality records, we fitted one LightGBM model per outcome on 2022 singleton births and evaluated preterm delivery, neonatal intensive care unit admission, and low birth weight on a 2023 temporal test. Early eligibility, defined as prenatal-care initiation in months 1–3, covered approximately 74.7% of records but only 69.4–71.8% of outcome events. At an equal absolute referral budget of 10% of each target-specific population, the early policy captured 2.58, 2.73, and 3.43 percentage points fewer events than the full-cohort policy, corresponding to 7,821, 8,271, and 8,356 events. Paired full-size bootstrap intervals from 500 replicates were above zero for all outcomes. Clinical prediction evaluation should therefore distinguish scoreability ceiling, conditional recall, and population event capture under an explicit resource budget.

Keywords: clinical risk prediction; cohort eligibility; scoreability; prenatal care; resource-constrained triage; temporal evaluation

Data and Code Availability The study uses publicly available, de-identified CDC Natality public-use records. The review package will include anonymized preprocessing and analysis code, locked configurations, input hashes, and article-facing result tables. A de-anonymized repository will be released after review.

Institutional Review Board (IRB) The analysis used publicly available, de-identified public-use records and involved no participant interaction or access to identifiable information. The applicable institutional determination will be provided if the paper is accepted.

1. Introduction

Clinical prediction can fail at two stages. A person may be unavailable for scoring because the required encounter or measurement has not occurred, or a model may rank an available case incorrectly. Standard discrimination and within-cohort recall primarily address the second stage and do not count events outside the scoreable cohort.

This distinction is consequential in prenatal triage. A policy restricted to people initiating prenatal care early cannot act on later-entry, no-care, or unknown-timing records. Conventional top- k comparisons can also confound eligibility with capacity when k is defined separately in cohorts of different sizes. We therefore compare early-entry and full-cohort policies while holding the fitted model, prediction vector, and absolute referral budget fixed.

Prior work treats nonrandom observation, post-score abstention, and finite clinical capacity largely as separate problems. Informative-presence studies show that observed EHR cohorts can differ systematically from target populations (??). Selective-prediction methods abstain after the required inputs reach the model (?), whereas resource-constrained evaluation can impose an absolute capacity among already scored candidates (?). Prenatal studies have used gestational landmarks or national vital-statistics cohorts (???), but generally report performance within records meeting each study’s inclusion and data-availability criteria. In our targeted audit, we found no study that retained events outside the scoreable cohort in the population denominator while fixing both the model and the absolute referral budget.

2. Methods

Data and population. CDC Natality 2022 contained 3,676,029 records and was used for model development and internal held-out evaluation. CDC 2023 contained 3,605,081 records and was used only

Figure 1: Fixed-model, equal-budget study design. One outcome-specific LightGBM model is fitted using CDC 2022, applied once to the CDC 2023 temporal test, and evaluated under early-entry and full-cohort eligibility. Both policies select the same absolute number of records. Events in groups unavailable to the early policy remain in the full-population denominator.

as a temporal test. The primary population comprised U.S.-resident singleton births with a known outcome value.

Outcomes and eligibility. Outcomes were preterm delivery (obstetric-estimate gestational age below 37 weeks), neonatal intensive care unit (NICU) admission, and low birth weight (LBW; below 2,500 g). Early eligibility required prenatal-care initiation in months 1–3. Later entry, no prenatal care, and unknown timing remained in the full-population denominator but were unavailable to the early policy. Source code 99 and blank prenatal-care-month values were mapped to missing; 0 was retained as no prenatal care.

Fixed-model comparison. For each outcome, one LightGBM model was fitted on CDC 2022 using the pre-specified `booking-strict` features (350 trees, learning rate 0.05, 31 leaves, and `min_child_samples=100`). We used no class weighting, oversampling, or hyperparameter optimization. Both policies used the same fitted model and the same CDC 2023 prediction vector.

For target-specific full-population size N_{full} , the absolute budget was

$$B = \text{round}(qN_{\text{full}}), \quad q \in \{0.05, 0.10\}.$$

Each policy selected the top B risks within its eligible set. Let E_{full} be all population events, E_{early} events in the early cohort, and $\text{TP}_s(B)$ events captured by policy s . We report

$$\begin{aligned} \text{Ceiling} &= \frac{E_{\text{early}}}{E_{\text{full}}}, \\ \text{Recall}_s &= \frac{\text{TP}_s(B)}{E_s}, \\ \text{Capture}_s &= \frac{\text{TP}_s(B)}{E_{\text{full}}}. \end{aligned}$$

The primary estimand was full-minus-early population event capture.

Uncertainty and supporting analyses. We used 500 paired, full-size, record-level bootstrap replicates with identical multiplicities for both policies and reconstructed top- B selection in every replicate. Percentile intervals were computed for the full-minus-early difference. At the 10% budget, true positives were decomposed by care-entry group. Platt calibration was fitted on CDC 2022 and evaluated on held-out 2022 and temporal-test 2023 data. We also repeated the analysis among all U.S.-resident births; this remained sensitivity evidence because pregnancy-level clustering could not be reconstructed.

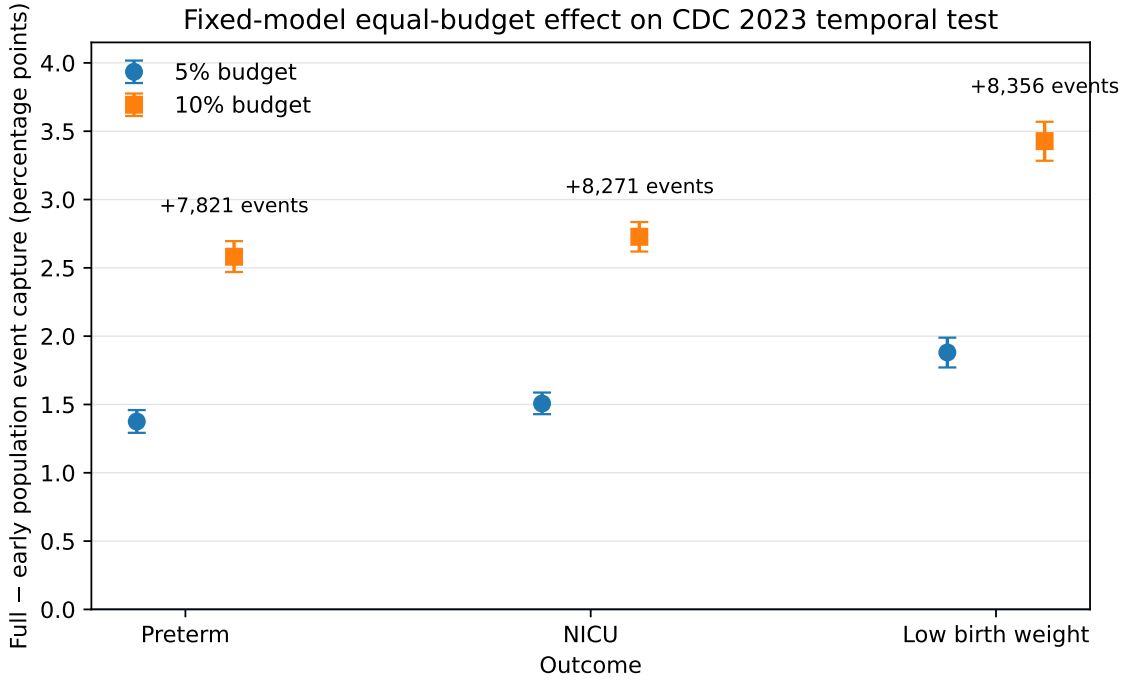
3. Results

Availability imposed an event ceiling. Early eligibility covered 74.71–74.73% of CDC 2023 records but only 71.83% of preterm events, 70.62% of NICU admissions, and 69.39% of LBW events (Table ??). Thus, 28–31% of events were outside the early-scoreable cohort before classifier error was considered.

Full eligibility increased capture at equal capacity. At the 10% budget, full-cohort capture exceeded early-entry capture by 2.58, 2.73, and 3.43 percentage points for preterm delivery, NICU admission, and LBW, representing 7,821, 8,271, and 8,356 additional events. At 5%, gains were 1.38, 1.51, and 1.88 points (Figure ??). Full precision was also higher at the equal 10% budget: 18.64% versus 16.39% for preterm delivery, 17.88% versus 15.50% for NICU admission, and 16.80% versus 14.40% for LBW.

Uncertainty was consistently positive. All CDC 2023 intervals were above zero. At 10%, intervals were 2.47–2.70 points for preterm delivery, 2.62–2.84 for NICU admission, and 3.28–3.57 for LBW. Held-out CDC 2022 produced similar differences of 2.60, 2.59, and 3.12 points, respectively, with intervals of 2.26–2.98, 2.29–2.95, and 2.65–3.62 points.

Gains arose outside the early cohort. At 10%, full eligibility gained 22,271 preterm, 20,557 NICU,



Paired full-size record-level bootstrap; 500 replicates. Early and full policies used the same target-specific absolute budget.

Figure 2: Population event capture under equal absolute referral budgets. Circles show early-entry eligibility and squares show full-cohort eligibility. Connecting segments show the change in captured population events; labels report the full-minus-early difference with its paired 95% bootstrap interval and the number of additional events.

and 21,857 LBW true positives from later-entry, no-care, and unknown-timing groups. The same budget reallocation reduced true positives within months 1–3 by 14,450, 12,286, and 13,501, producing the net gains above.

Supporting analyses were consistent. Raw CDC 2023 full-cohort models had calibration intercepts from -0.019 to 0.015 and slopes from 0.995 to 0.998 ; Platt scaling did not improve Brier score. All-birth 10% differences remained positive at 2.18, 2.51, and 2.85 points for preterm delivery, NICU admission, and LBW.

4. Discussion

Restricting scoring to prenatal-care entry in months 1–3 reduced population capture after fixing the model, predictions, and absolute budget. The result

is not a claim about a superior architecture: eligibility determines which events are available for selection. It also differs from selective prediction, which abstains after a scoreable case reaches the model. Here, events outside the early cohort remain in the denominator because they remain part of the population the triage policy is intended to serve.

The mechanism analysis showed the operational trade-off. Full eligibility reallocated the same referral capacity from some early-entry records toward high-risk later-entry, no-care, and unknown-timing records. Gains outside the early cohort exceeded losses within it for all outcomes.

Limitations include retrospective administrative data, non-optimized 5% and 10% capacity scenarios, temporal rather than cross-system evaluation, and no causal interpretation of care entry. Multiple-birth records may be dependent, and pregnancy-level

Table 1: CDC 2023 temporal-test results at the target-specific 10% absolute referral budget. Capture is population event capture; intervals are paired bootstrap percentile intervals from 500 replicates.

Outcome	Full <i>N</i>	Early <i>N</i> (%)	Events <i>n</i> (%)	Event ceiling	Budget <i>B</i>	Early capture	Full capture	Δ capture, pp (95% CI)	Additional TP
Preterm delivery	3,480,382	2,601,060 (74.73)	303,001 (8.71)	71.83	348,038	18.83	21.41	2.58 (2.47–2.70)	7,821
NICU admission	3,476,584	2,597,274 (74.71)	303,285 (8.72)	70.62	347,658	17.77	20.50	2.73 (2.62–2.84)	8,271
Low birth weight	3,480,316	2,600,093 (74.71)	243,776 (7.00)	69.39	348,032	20.56	23.98	3.43 (3.28–3.57)	8,356

clustered uncertainty was unavailable. Secondary **pregnancy_oracle** variables were not reliably available at the early clinical landmark and are not presented as deployable inputs.

Clinical triage evaluations should separately report scoreability ceiling, conditional performance among eligible records, and population event capture under an explicit absolute budget.

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